

```
/*  
 * C program to perform quick sort on a user provided array using recursion  
 */
```

```
#include <stdio.h>
```

```
int partition(int ar[], int low, int high)
```

```
{  
    int pivot = ar[high];  
    int i = low - 1;  
    for (int j = low; j < high; j++)  
    {  
        if (ar[j] <= pivot)  
        {  
            i++;  
            int temp = ar[i];  
            ar[i] = ar[j];  
            ar[j] = temp;  
        }  
    }  
    int temp = ar[i + 1];  
    ar[i + 1] = ar[high];  
    ar[high] = temp;  
    return i + 1;
```

```
}
```

```
void quickSort(int arr[], int low, int high)
```

```
{
```

```
    if (low < high)
```

```
    {
```

```
        int pi = partition(arr, low, high);
```

```
        quickSort(arr, low, pi - 1);
```

```
        quickSort(arr, pi + 1, high);
```

```
    }
```

```
}
```

```
int main(void)
```

```
{
```

```
    puts("Enter the number of elements in the array: ");
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    int arr[n];
```

```
    puts("Enter the elements of the array: ");
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        printf("%d: ", i);
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
quickSort(arr, 0, n - 1);  
  
puts("The sorted array is: ");  
  
for (int i = 0; i < n; i++)  
{  
    printf("%d ", arr[i]);  
}  
}
```